
Python Basics Cheatsheet

Scientific Programming

1. Basic Calculation Operations

```
5 + 3 # 8          5 / 3 # 1.66...
5 - 3 # 2          5 // 3 # 1
5 * 3 # 15         5 % 3 # 2
5 ** 3 # 125
```

Rounding

```
round(1.6667) # 2
round(1.6667, 2) # 1.67
```

2. Math (not allowed during exam!)

Import the math library
`import math`

Constants

```
math.pi # 3.14159...
math.e # 2.718...
```

Power and Roots

```
math.sqrt(16) # 4
math.pow(2, 3) # 8
math.exp(1) # e^1
math.log(10) # log(10) base e
math.log(100, 10) # log(10) base 10
```

Trigonometric Functions

```
math.sin(math.pi / 2) # 1
math.cos(0) # 1
math.tan(math.pi / 4) # 1
```

Inverse Trigonometric Functions

```
math.asin(1) # pi/2
math.acos(1) # 0
math.atan(1) # pi/4
```

Rounding

```
math.floor(3.7) # 3
math.ceil(3.2) # 4
```

3. Basic Type Conversions

```
int("42") # String to int
float("3.14") # String to float
str(42) # Int to string
list("abcd") # String to list
```

4. Lists

Create a list

```
my_list = [3, 2, 1]
```

Operations

```
item = my_list[2] # 1 - element at index 2
my_list.append(3) # [3, 2, 1, 3]
new_list = my_list + [5, 6] # [3, 2, 1, 3, 5, 6]
x = len(new_list) # 6 - length of list
"Apple" in new_list # False
```

Looping 1

```
for item in my_list:
    print(item)
```

Looping 2

```
for i in range(len(my_list)):
    item = my_list[i]
    print(item)
```

Methods

```
new_list.sort() # [1, 2, 3, 3, 5, 6]
new_list.insert(2, 10) # [1, 2, 10, 3, 3, 5, 6]
new_list.remove(3) # [1, 2, 10, 3, 5, 6]
i = new_list.index(10) # 2 (↑ index of 10: 2)
```

Slicing

```
lst = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
lst[2:5] # [2, 3, 4] (from 2 to 4)
lst[:3] # [0, 1, 2] (from start to 2)
lst[7:] # [7, 8, 9] (from 7 to end)
lst[-3:] # [7, 8, 9] (last 3)
lst[::2] # [0, 2, 4, 6, 8] (every 2nd)
lst[1:8:3] # [1, 4, 7] (from 1 to 7 step 3)
lst[::-1] # [9, 8, ...] (reverses list)
lst[8:2:-2] # [8, 6, 4] (from 8 to 3 step -2)
```

5. Strings

Strings

```
my_str = "Hello, World!"
```

Methods

```
my_str.lower() # Convert to lowercase
my_str.upper() # Convert to uppercase
my_str.lstrip() # Remove leading whitespace
my_str.rstrip() # Remove trailing whitespace
my_str.strip() # Combined lstrip and rstrip
my_str.isalpha() # all characters alphabetic?
my_str.isdigit() # all characters digits?
my_str.islower() # all characters lowercase?
my_str.isupper() # all characters uppercase?
my_str.split(" ") # split string on spaces
```

Looping through characters 1

```
for char in my_str:
    print(char)
```

Looping through characters 2

```
for i in range(len(my_str)):
    char = my_str[i]
    print(char)
```

6. Reading files

Open and read whole file

```
file = open("file.txt", "r")
content = file.read()
print(content)
file.close()
```

Open and read line by line

```
file = open("file.txt", "r")
for line in file:
    print(line)
file.close()
```

Alternative method: with open

```
with open("file.txt", "r") as file:
    for line in file:
        print(line)
```
